

Human African Trypanosomiasis

Progress and Stagnation



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KEYWORDS

- Human African trypanosomiasis • *Trypanosoma brucei gambiense*
- *Trypanosoma brucei rhodesiense* • Clinical presentation • Diagnosis • Treatment

KEY POINTS

- Human African trypanosomiasis (HAT) is caused by 2 subspecies of the protozoon *Trypanosoma brucei*, *T brucei gambiense*, and *T brucei rhodesiense*, that are transmitted by tsetse flies in sub-Saharan Africa.
- The clinical course of both infections is rather different, with *T brucei gambiense* causing a chronic illness affecting predominantly the central nervous system of patients living in endemic areas, and *T brucei rhodesiense* mainly presenting as a more acute systemic febrile illness in local populations and in travelers.
- The burden of *T brucei gambiense* HAT is rapidly decreasing in Central and West Africa for 15 years, whereas *T brucei rhodesiense* infections continue to be sporadically observed in local populations as well as in travelers exposed in game reserves in East Africa.
- New antibody-based rapid diagnostic tests have the potential to simplify and accelerate the diagnosis of *T brucei gambiense* HAT in the field. Microscopy remains the main method to diagnose *T brucei rhodesiense* infection.
- Field treatment of second-stage *T brucei gambiense* HAT has evolved from highly toxic melarsoprol regimens to a safer and more convenient combination of intravenous eflornithine and oral nifurtimox, which remains, however, cumbersome for most rural settings in the tropics. Fexinidazole, a new oral treatment, has recently shown clinical noninferiority in the treatment of the second stage of the disease, and has therefore the potential to revolutionize HAT control. No progress has been achieved for the therapy of second-stage *T brucei rhodesiense* HAT, for which melarsoprol remains the only widely accepted option.

INTRODUCTION

Human African trypanosomiasis (HAT), also called sleeping sickness, is probably one of the most exemplary tropical diseases, because its vector, the blood-feeding tsetse

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fly (Diptera, genus *Glossina*), is exclusively distributed in intertropical Africa. It is a disease complex caused by 2 different subspecies of *Trypanosoma brucei*: *Trypanosoma brucei gambiense* and *Trypanosoma brucei rhodesiense*, which are each transmitted by different species or subspecies of *Glossina*. Despite their undistinguishable aspect under a microscope and similarities in transmission and disease evolution, both pathogens widely differ in terms of epidemiology, clinical manifestations, diagnostic approach, and response to anti-parasitic treatment. *T. brucei gambiense* is by far the most prevalent subspecies and causes the anthroponotic West African HAT, which has been associated with major epidemic waves in the twentieth century. In contrast, *T. brucei rhodesiense* is a zoonotic pathogen causing East African HAT that sporadically affects humans. The burden of both *T. brucei gambiense* and *T. brucei rhodesiense* HAT has substantially decreased for 15 years in the endemic areas, thanks to intense control efforts.¹ HAT cases are, however, occasionally seen outside Africa, where they often pose significant management challenges to physicians not familiar with tropical medicine.² The purpose of this review was to provide an update on the recent advances in the diagnosis and treatment of both forms of HAT, with a focus on the differing aspects and perspectives of their clinical management in the nonendemic setting.

CURRENT EPIDEMIOLOGY IN ENDEMIC COUNTRIES AND IN TRAVELERS

The distribution of HAT is highly focal, resulting in complex interactions among parasite, vector, host, and environment. Transmission occurs in rural or periurban areas when humans are exposed to the bites of forest, riverine, or savannah tsetse flies.¹ Congenital transmission is possible,^{3,4} but other routes, such as laboratory accidents, blood transfusion, or organ transplantation are likely extremely rare.

There are approximately 360 historical foci of HAT distributed in 36 sub-Saharan Africa countries: 300 foci of *T. brucei gambiense* HAT in 24 West and Central African countries and 60 foci of *T. brucei rhodesiense* HAT in 13 East African countries; one country, Uganda, is endemic for both infections that do not regionally overlap.⁵ Nowadays, approximately 70 million people live at different levels of risk for HAT infection. The Atlas of Human African Trypanosomiasis⁶ provides dynamic and regularly updated information from 2000 about the number of HAT cases at the village level (accessible at http://www.who.int/trypanosomiasis_african/country/risk_AFRO/en/). *T. brucei gambiense* HAT accounts for 97% of the total burden. No cases of HAT have been reported since 2010 in 15 of the countries considered as endemic.⁷

***Trypanosoma brucei gambiense* Human African Trypanosomiasis Is Contemplated for Elimination**

Compared with the alarming levels observed in early 2000 after a long period of neglect, the total number of HAT cases has dramatically decreased to a record low number of 2164 in 2016, corresponding to a 92% reduction compared with this period (http://www.who.int/trypanosomiasis_african/news/The-world-gears-up-to-eliminate-HAT-2020/en/, accessed on June 6, 2018). This success has been made possible through major international and national reinvestments in active mass screening, vector control activities, and improved case management.⁸ Currently, risk remains moderate to high (>1/10,000 inhabitants per year) for approximately 10 million people in areas where HAT is still considered a public health problem, mainly in the Democratic Republic of the Congo.⁷ Recently, HAT has been included in the World Health Organization (WHO) Neglected Tropical Disease roadmap as one of the diseases targeted for elimination as a public health problem by 2020. To achieve this goal, however, the role of

cryptic reservoirs such as latent human infections or possible animal reservoirs in *T brucei gambiense* HAT transmission still needs to be clarified.⁹

***Trypanosoma brucei rhodesiense* Human African Trypanosomiasis Remains a Persistent Threat in International Travelers**

The number of *T brucei rhodesiense* HAT cases has also decreased since 2000, and moderate risk of infection is now limited to restricted areas of Uganda, Malawi, and Zambia. Progress seems to stagnate since 2011, with approximately 100 cases diagnosed per year in endemic countries.⁷

In travelers, epidemiology of HAT has evolved in the past century. In early 1900, *T brucei gambiense* HAT was predominant in the nonendemic setting and mainly seen in expatriates and soldiers.¹⁰ For the past 25 years, the number of published cases has increased above 100, with a vast majority of *T brucei rhodesiense* infections diagnosed in all continents, including Asia recently, among short-term tourists who had visited national parks and game reserves in East Africa.^{8–24} In comparison, during this period, fewer than 30 cases of *T brucei gambiense* infections have been reported outside Africa, almost exclusively in migrants or long-term expatriates, sometimes from foci considered as silent,¹² or after a very long incubation period of up to 30 years.²⁵ HAT accounted for 6 (0.06%) of the 10,092 international travelers presenting with systemic febrile illness in the last survey of the GeoSentinel network of travel clinics from 2007 to 2011.²⁶

CLINICAL AND LABORATORY FEATURES

Both forms of HAT progress from skin infection at the site of a tsetse bite (inoculation chancre), caused by local trypanosome multiplication, to systemic symptoms following dissemination in the bloodstream and in the reticuloendothelial system (first, or early, or hemolymphatic stage). Infection gradually progresses with invasion of the central nervous system (CNS) when the parasites cross the blood-brain barrier (second, or late, or meningoencephalitic stage). The main difference, however, resides in the speed of parasite multiplication and the rapidity of symptom evolution. Systemic and neurologic symptoms may overlap, so that the distinction between first and second stage cannot rely on clinical presentation, but needs to be based on stringent laboratory criteria (**Box 1** and section on “Diagnosis”).

Table 1 summarizes the approximate frequencies of the main clinical features per trypanosome subspecies and disease stage, as reported in recent series from endemic settings.²⁷ Most studies described, however, mixed groups of patients with first- and second-stage disease, and only a few small case series focused on first-stage only, probably because both forms of HAT are disproportionally diagnosed at the late stage in the field.

Box 1

Criteria to confirm second-stage human African trypanosomiasis

Demonstration of trypanosomes in any body fluid (blood, cervical lymph node aspirate, or cerebrospinal fluid) AND

Cerebrospinal fluid examination demonstrating:

- The presence of trypanosomes,
- And/or the presence of more than 5 white blood cells per μL

From World Health Organization. Control and surveillance of human African trypanosomiasis. World Health Organ Tech Rep Ser 2013;(984); with permission.

Clinical and Laboratory Features	First-Stage <i>T.b. gambiense</i> HAT	Second-Stage <i>T.b. gambiense</i> HAT	First-Stage <i>T.b. rhodesiense</i> HAT	First-Stage <i>T.b. rhodesiense</i> HAT	Second-Stage <i>T.b. rhodesiense</i> HAT
N evaluated	39 ²⁹ + 56 ³⁰	2541 ³¹	83 ³⁸	45 ³⁶	138 ³⁹ + 192 ³⁸
Setting	Endemic	Endemic	Endemic	Travelers	Endemic
Non-neurological features					
Chancre	<5	0	50	85	15
Documented fever	20	15	50	100	20
Headache	80	80	70	50	95
Anorexia	—	25	—	—	85
Pruritus	45	50	—	5	20
Lymphadenopathy	60	55	50	40	30
Any edema (face,...)	15	35	25	—	40
Jaundice	—	—	—	25	—
Skin rash	—	—	—	30	—
Splenomegaly	50	15	15	25	25
Neurologic features					
Any sleeping disorder	60	75	25	10	85
Daytime sleeping	—	40	25	5	75
Insomnia	—	55	—	5	65
Behavior change	10	30	—	5	20
Motor weakness	—	35	—	—	20
Walking difficulties	—	20	—	—	50
Altered consciousness	—	15	5	<5	20
Sensory-motor deficit	—	35	—	—	20
Tremor	—	20	30	5	40
Abnormal movements	15	10	—	<5	25
Speech impairment	—	15	—	—	15
Laboratory findings					
CSF-WBC 6–19/μL	—	22	—	—	25
CSF-WBC 20–99/μL	—	32	—	—	38
CSF-WBC >100/μL	—	45	—	—	37
Trypanosomes in CSF	—	40	—	—	85
Trypanosomes in blood	30–75	30–75	95	95	95

Abbreviations: CSF, cerebrospinal fluid; HAT, human African trypanosomiasis; *T.b.*, *Trypanosoma brucei*; WBC, white blood cell.

From World Health Organization. Control and surveillance of human African trypanosomiasis. World Health Organ Tech Rep Ser 2013;(984); with permission.

***Trypanosoma brucei gambiense* Human African Trypanosomiasis Mainly Presents as a Slow-Progressing Neurologic Disease**

The trypanosomal chancre often goes unrecognized. The first stage, which lasts for approximately 1 year on average,²⁸ is characterized by intermittent low-grade fever;

nonspecific symptoms, such as headache, anorexia, and asthenia; painless lymphadenopathies with preferential cervical localization (Winterbottom sign); pruritus; and some subtle neurologic manifestations (sleep disorders, concentration disturbances).^{29,30} Clinical information on the second stage is much more robust.³¹ It presents with more prominent neurologic and neuropsychiatric manifestations, while most systemic symptoms tend to regress. The neurologic functions deteriorate progressively but ineluctably, causing a wide variety of symptoms and signs, which may include nocturnal insomnia, daytime hypersomnia, tremor, fasciculations, ataxia, motor weakness, archaic reflexes, abnormal behaviors, confusion, coma, or dementia.³² If left untreated, death occurs on average from 1 to 2 years after neurologic involvement begins.³³ Subclinical myocarditis and endocrine hypothalamic-hypophyseal dysfunction are often present.^{1,34,35}

In the nonendemic setting, the clinical presentation of the few reported *T brucei gambiense* HAT cases differed between Western travelers and African migrants.³⁶ Although the latter were almost exclusively diagnosed during the late stage of the disease,¹⁷ with pattern and frequency of symptoms similar to those described previously, many travelers and expatriates presented with early-stage symptoms (eg, chancre, fever or rash) almost identical to those of *T brucei rhodesiense* infection (see the following section).

***Trypanosoma brucei rhodesiense* Human African Trypanosomiasis Is Mainly an Acute Systemic Febrile Illness**

The trypanosomal inoculation chancre is often still present at the onset of systemic symptoms, such as fever, generalized tender lymphadenopathies, splenomegaly, and/or facial or generalized edema.^{37,38} At that stage, parasitemia is usually high (see section of “Diagnosis”). Multiorgan failure and acute myocarditis may develop at any time during this period. Neurologic involvement may appear a few weeks later, presenting as subacute meningoencephalopathy,³⁹ and death usually occurs within 6 months after the initial onset of symptoms. Recent studies have shown, however, that the clinical phenotypes may dramatically differ between endemic foci, even within a single country,³⁸ likely due to the genetic diversity of *T brucei rhodesiense* strains. Also, the reported symptoms and signs may be biased by the high frequency of coinfections, such as malaria, helminthic diseases, or human immunodeficiency virus (HIV) in endemic areas.³⁷

In the nonendemic setting, meaningful data on symptom frequencies have been collected for the early stage of *T brucei rhodesiense* HAT (see [Table 1](#)),³⁶ whereas only 10 cases with late-stage disease have been reported.^{10,11,17,40} In the largest review describing 45 travelers diagnosed with first-stage *T brucei rhodesiense* HAT,³⁶ the trypanosomal chancre was almost always present at diagnosis, at the time of systemic symptoms that usually appear 3 to 14 days after the infective bite. Rash and jaundice were frequently observed, in 30% and 25% of the cases, respectively, whereas these features have never been reported in series from endemic countries. Symptoms of encephalopathy were present in a minority of travelers with early-stage disease. In almost all cases, thrombocytopenia, elevated levels of inflammatory parameters, liver enzymes, and bilirubin, as well as kidney dysfunction, were observed. In contrast, white blood cell count was often normal.

Of note, in travelers who have been exposed in different regions where HAT occurs, the similarities in the clinical presentation of *T brucei rhodesiense* and *T brucei gambiense* HAT could pose a serious diagnostic dilemma.^{17,36}

DIAGNOSIS

Because current treatments of HAT, and in particular of its late stage, are complex and potentially toxic, all efforts should be made to first obtain a diagnosis of certainty, by demonstrating the presence of trypanosomes in any body fluid. Once the diagnosis of infection is established, the second step is to stage the disease, by looking in the cerebrospinal fluid (CSF) for trypanosomes and for indirect markers of CNS involvement.²⁷ The set of operational criteria that has been established by WHO to ascertain the diagnosis of second-stage HAT in endemic settings is summarized in **Box 1**. Direct identification of trypanosomes in CSF is poorly sensitive (see **Table 1**). Although the cutoff of 5 white blood cells (WBCs) per μL of CSF has often been challenged, it is the most widely accepted for clinical management, as far as the counting procedure is repeated for WBC counts between 5 and $20/\mu\text{L}$, and that the average of both measures is taken as the final result.⁴¹ Determination of protein levels in CSF is not systematically performed in the field and seems to have less value. Elevated concentration of immunoglobulin (Ig)M in CSF is a very sensitive marker of second-stage infection, but is not widely available.⁴² Novel biomarkers are being evaluated to improve the diagnostic accuracy of second-stage HAT.^{43,44}

Parasitologic Diagnosis of Trypanosoma brucei gambiense Human African Trypanosomiasis Is Notoriously Difficult

Diagnosis may be obtained by direct microscopic examination of mobile trypanosomes in blood, or after Giemsa staining (**Table 2**). Both techniques have a low diagnostic yield, even when repeated (30%). Direct examination of cervical lymph node aspirate or CSF is also poorly sensitive.^{45,46} Sensitivity may be substantially improved by concentration methods that require additional laboratory infrastructure and skills not always available in endemic settings. The techniques most widely used are the microhematocrit or capillary tube centrifugation (mHTC or CTC or Woo method) and the mini-anion exchange centrifugation technique (mAECT, or “mini-column”), that increase sensitivity to 60% and 80%, respectively.^{45,46} Subgenus-specific polymerase chain reaction (PCR) assays mainly targeting the 177-bp satellite DNA have been recently developed and shown high sensitivity and specificity (see **Table 2**). They are, however, for the moment available only in reference laboratories. New low-tech molecular methods such as loop-mediated isothermal amplification have been designed but are not readily available in most HAT settings.^{46–48} The most interesting next-generation diagnostic for active infection by trypanosomatids is the spliced leader RNA (SL-RNA) detected by PCR.⁴⁹

Because parasitologic diagnosis is cumbersome and poorly sensitive, antibody-based serologic assays have been developed for >50 years as screening tools. The most widely used in the field is the Card Agglutination Test for Trypanosomiasis (CATT/T *b. gambiense*), developed at the Institute of Tropical Medicine, Antwerp (ITMA), which has been the cornerstone of mass screening campaigns for years. Sensitivity and specificity were high enough ($\geq 95\%$ in most but not all HAT-endemic settings) to make it an excellent field tool to select individuals requiring additional confirmatory testing (with conventional microscopy and concentration methods).⁴⁵ Because the CATT is designed for mass screening and still requires agitator rotator, electricity, and refrigeration, there is a need for simple and individual point-of-care tests. Rapid diagnostic tests (RDTs) based on immunochromatographic lateral-flow detection of antibodies against the variant surface glycoproteins (VSGs) LiTat 1.3 and VSG LiTat 1.5 have been recently developed and evaluated.⁵⁰ The lateral-flow device HAT Sero-K-SeT test, developed by Coris BioConcept (Gembloux,

Table 2
Diagnostic accuracy of the different laboratory methods for the diagnosis of *Trypanosoma brucei gambiense* human African trypanosomiasis (HAT)

	Threshold of Detection	Sensitivity	Specificity
Microscopy-based methods			
Examination of wet blood	10,000/mL	5–55	100
Blood film (Giemsa staining)	—	25–50	100
Examination lymph node aspirate	—	20–60	100
Capillary tube centrifugation (or Woo)	500/mL	45–90	100
Mini-anion-exchange centrifugation technique	50/mL	75–90	100
Molecular methods			
Polymerase chain reaction assays	1–1000/mL	87–100	92–100
DNA-based loop-mediated isothermal amplification	100–1000/mL	75	100
Spliced leader RNA	100/mL	92	96
Antibody-based methods			
Card Agglutination Test for Trypanosomiasis	—	87–98	95
HAT Sero-K-SeT (Coris BioConcept, Gembloux, Belgium)	—	98–100	97–98
SD BIOLINE HAT (Standard Diagnostics, Kyonggi-do, South Korea)	—	89	95
SD BIOLINE HAT 2.0 (Standard Diagnostics, Kyonggi-do, South Korea)	—	70	98

Adapted from Chappuis F, Loutan L, Simarro P, et al. Options for field diagnosis of human African trypanosomiasis. Clin Microbiol Rev 2005;18(1):133–46; and Mitashi P, Hasker E, Lejon V, et al. Human African Trypanosomiasis diagnosis in first-line health services of endemic countries, a systematic review. PLoS Negl Trop Dis 2012;6(11); with permission.

Belgium) in collaboration with the ITMA, displayed sensitivity and specificity $\geq 98\%$ in a diagnostic study on stored serum samples from *T. brucei gambiense* HAT cases and healthy controls.⁵¹ Subsequent prospective diagnostic studies in endemic foci⁵² and clinical suspects⁵³ confirmed this excellent accuracy, with sensitivity and specificity of 98% and 98% and of 100% and 97%, respectively, comparable to those of CATT. A second HAT-RDT has been developed by Alere/Standard Diagnostics (SD; Kyonggi-do, South Korea) in partnership with the Foundation for Innovative New Diagnostics, first with native VSG antigens (SD BIOLINE HAT) and second with recombinant antigens (SD BIOLINE HAT 2.0). In large multicenter prospective studies, sensitivity of both SD HAT-RDTs was found lower than expected (71%–89%) whereas specificity was very high (98%).^{54,55} However, combining any of these RDTs together or with CATT achieved a very high sensitivity. It appears therefore that both these RDTs achieve a diagnostic accuracy equivalent to that of CATT and may be used instead for both mass screening and clinical care, wherever local conditions do not favor the use of CATT.

In travelers, as in endemic settings, only a combination of microscopy-based and antibody-based methods may reach adequate sensitivity, whereas molecular assays

could be of added value. This difficulty explains the diagnostic delay in patients with *T brucei gambiense* infection being explored outside Africa.³⁶ All methods require, however, expertise and material that is available in only a few reference laboratories, such as the ITMA in Antwerp, Belgium, or the Centers for Disease Control and Prevention, Atlanta, GA.

Diagnosis of Acute Trypanosoma brucei rhodesiense Human African Trypanosomiasis Is Straightforward

Because the parasite load in blood is usually very high at symptom onset, trypanosomes are relatively easily detected whenever a blood smear is performed, as it is common practice in a febrile traveler after a stay in Africa. In travelers, diagnosis of *T brucei rhodesiense* HAT has been almost always made by thick and thin blood film examination, often as a surprise finding.^{17,36} It could, however, be missed when in such circumstances malaria diagnosis is limited to RDT. Although antibody-based assays exist for *T brucei rhodesiense* HAT, none have been developed in RDT format.

In travelers with multiple exposure to different HAT-endemic regions, species-specific diagnosis is important to select the most appropriate treatment. Molecular techniques for species differentiation are available in only a few reference settings.⁴⁷

TREATMENT

Selection of HAT therapy still relies on the recognition of the causative trypanosome and on disease staging. To treat the hemolymphatic phase, 2 drugs have been used for decades: pentamidine and suramin. Both drugs do not penetrate the CNS. For the meningoencephalitic stage, melarsoprol, eflornithine, and the nifurtimox-eflornithine combination therapy (NECT) are the drugs currently used, but 2 new drugs, fexinidazole and acoziborole, offer promising perspectives. The current treatment recommendations for endemic and nonendemic settings and the main adverse events are presented in [Table 3](#).

Trypanosoma brucei gambiense Human African Trypanosomiasis Treatment Is Being Revolutionized

Hemolymphatic stage

Pentamidine isethionate for 7 days remains the treatment of choice for first-stage infection due to *T brucei gambiense*, because it is rather well tolerated and remarkably effective (cure rate more than 93%) despite more than 50 years of widespread use throughout Africa.⁵⁶ Despite promising results in a phase 2 trial,⁵⁷ the development of panfuramide as an alternative to pentamidine was stopped in 2008 after safety concerns emerged during the phase 3 trial. Two participants developed reversible glomerulopathy.⁵⁸ Suramin has also some activity against *T brucei gambiense*, but its lower efficacy and higher toxicity preclude its use for the first-stage *T brucei gambiense* HAT. In contrast, preliminary results suggest that the new compound fexinidazole is highly effective in first-stage *T brucei gambiense* HAT (see later in this article).

Meningoencephalitic stage

Melarsoprol, an arsenical compound, has been used for more than 50 years for treatment of second-stage HAT. It has an excellent trypanocidal activity, compensating its rather low CNS penetration. However, toxicity has always been a major issue, in particular because of the encephalopathic syndrome occurring in up to 10% of the treated patients and associated with almost 50% mortality,⁵⁹ that could be only partly prevented by coadministration of prednisolone.⁶⁰ This complication seems either due

Table 3

Recommended treatment of human African trypanosomiasis in endemic settings and in travelers (and side effects per drug reported in >1% of the cases)

	First Choice	Second Choice
<i>Trypanosoma brucei gambiense</i>		
First-stage disease	Pentamidine 4 mg/kg per day IM or slow IV (diluted in saline in 2-h infusion) once-daily for 7 d. <i>Side effects: pain and swelling at the injection site, hypotension, abdominal discomfort, glucose instability and diabetes.</i>	Eflornithine 100 mg/kg 4 times per day (diluted in 250 mL saline for a 2-h infusion) for 14 d. NB: in children eflornithine 150 mg/kg 4 times per day (diluted in 250 mL saline for a 2-h infusion) for 14 d.
Second-stage disease	NECT: Eflornithine 200 mg/kg twice daily (diluted in 250 mL saline for a 2-h infusion) for 7 d combined with nifurtimox PO 15 mg/kg per day divided in 3 doses for 10 d. NB: in children eflornithine 300 mg/kg twice daily (diluted in 250 mL saline for a 2-h infusion) for 7 d combined with nifurtimox PO 15 mg/kg/d divided in 3 doses for 10 d. <i>Side effects eflornithine: headache, seizures, bone marrow toxicity.</i> <i>Side effects nifurtimox: headache, seizures, nausea, dizziness, tremor, skin rash.</i>	Eflornithine 100 mg/kg 4 times per day (diluted in 250 mL saline for a 2-h infusion) for 14 d. NB: first choice in pregnant women (nifurtimox contraindicated). NB: in children eflornithine 150 mg/kg 4 times a day (diluted in 250 mL saline for a 2-h infusion) for 14 d.
<i>Trypanosoma brucei rhodesiense</i>		
First-stage disease	Suramin: test dose of 100 mg IV (under monitoring) on day 1 followed by 20 mg/kg (max 1 g) on days 1, 3, 6, 14, 21 (or once a week for 5 wk). <i>Side effects: hypersensitivity, nephropathy, peripheral neuropathy and bone marrow toxicity.</i>	Pentamidine 4 mg/kg per day IM or slow IV (diluted in saline in 2-h infusion) once-daily for 7 d.
Second-stage disease	Melarsoprol: 2.2 mg/kg/d IV for 10 d (consider prednisolone 10 mg/kg preventively, or in case fever and convulsions develop after initiation of treatment). <i>Side effects: encephalopathic syndrome, melarsoprol reaction, hepatitis, pancreatitis, neuropathy, tremor, skin rash.</i>	

Abbreviations: IM, intramuscular; IV, intravenous; NECT, nifurtimox-eflornithine combination therapy; PO, by mouth.

From World Health Organization. Control and surveillance of human African trypanosomiasis. World Health Organ Tech Rep Ser 2013;(984); with permission.

to early direct cytotoxicity or late (7–10 days) antigen-mediated immune reaction related to parasite lysis. No symptoms or signs at presentation can predict the risk of developing an encephalitic syndrome, but some observations suggest that it was associated with the appearance of fever, rash, and convulsions after the start of the treatment.⁵⁹ Many other severe adverse events are possible in treated patients, such as arrhythmias, peripheral neuropathy, or hepatitis, reflecting multiorgan toxicity. In addition, treatment failure with melarsoprol was increasingly reported in many areas,⁶¹ possibly linked with mutations in the aquaglyceroporin 2 and 3 genes.⁶²

In the late 1980s, the polyamine synthesis inhibitor eflornithine (or difluoromethylornithine), emerged as an effective and much less toxic alternative drug to melarsoprol against second-stage *T brucei gambiense* HAT.^{63,64} However, because of its short half-life (1.5–5.0 hours) and its low binding with proteins, adequate efficacy could be achieved only with 4 daily intravenous administrations for a 2-week period. Shorter courses of eflornithine were clearly inferior.⁶⁵

Although manageable in well-equipped hospitals, this regimen was too cumbersome and expensive for the remote areas where the disease prevails, even with strong international support. Therefore, after pharmacologic exploration for synergism, the Drug for Neglected Diseases Initiative (DNDI) designed and conducted a pivotal randomized control trial that demonstrated that the combination of a 10-day course of oral nifurtimox with a simpler, 7-day twice-daily, intravenous course of eflornithine was not inferior to eflornithine monotherapy (cure rate more than 95%) and reduced major adverse events by half (14% vs 29%; $P = .002$).⁶⁶ Phase 4 field studies further confirmed that the NECT was safer and more convenient than, and as effective as, eflornithine monotherapy.^{67,68} This treatment has been endorsed by WHO since 2012 as first-line option for second-stage *T brucei gambiense* HAT. Children require higher dosages of eflornithine because of increased renal clearance. In pregnant women, eflornithine should be administered alone because nifurtimox is mutagenic. Treatment of HAT in immunosuppressed patients has never been specifically studied, but local experience suggests that HIV coinfection does not affect treatment outcome.

Very recently, another therapeutic breakthrough has been achieved by the DNDI, with the demonstration that an oral 10-day treatment with fexinidazole, a nitroimidazole analogue, is as safe as, and not inferior to, the NECT.⁶⁹ If confirmed in additional field studies, this new drug represents the first well-tolerated single-compound oral therapy against second-stage, and possibly also first-stage, *T brucei gambiense* HAT. Another drug, acoziborole (SCYX-7158), a single-dose oral oxaborole compound,⁷⁰ has recently entered a phase 2/3 DNDI-sponsored clinical trial for the treatment of first- and second-stage *T brucei gambiense* HAT. These new treatments could become a game changer in HAT treatment and control.⁷¹

The need of a long-term posttreatment follow-up to ascertain definitive cure has always been a particular challenge in endemic areas. Researchers have demonstrated that the follow-up strategy of second-stage *T brucei gambiense* HAT could be shortened to 1 year maximum, instead of the previous recommendation of CSF analysis every 6 months over 2 years. Algorithms based on CSF examination have been proposed to identify definitive cure at 6 months posttreatment (no trypanosome and ≤ 5 WBC/ μ L), or treatment failures (presence of trypanosomes and/or more than 50 WBC/ μ L). Patients with 6 to 49 WBC/ μ L at 6 months need further follow-up at 12 months (Fig. 1). Treatment failure is proven when trypanosomes are seen and likely when CSF microscopy reveals more than 20 WBC/ μ L.⁷² CSF examination 3 months after treatment does not provide reliable information. Novel biological predictors for treatment failure have been developed.^{73–75} In contrast, the CATT antibody test or detection of trypanosome DNA by PCR is not appropriate for evaluation of treatment response.⁷⁶

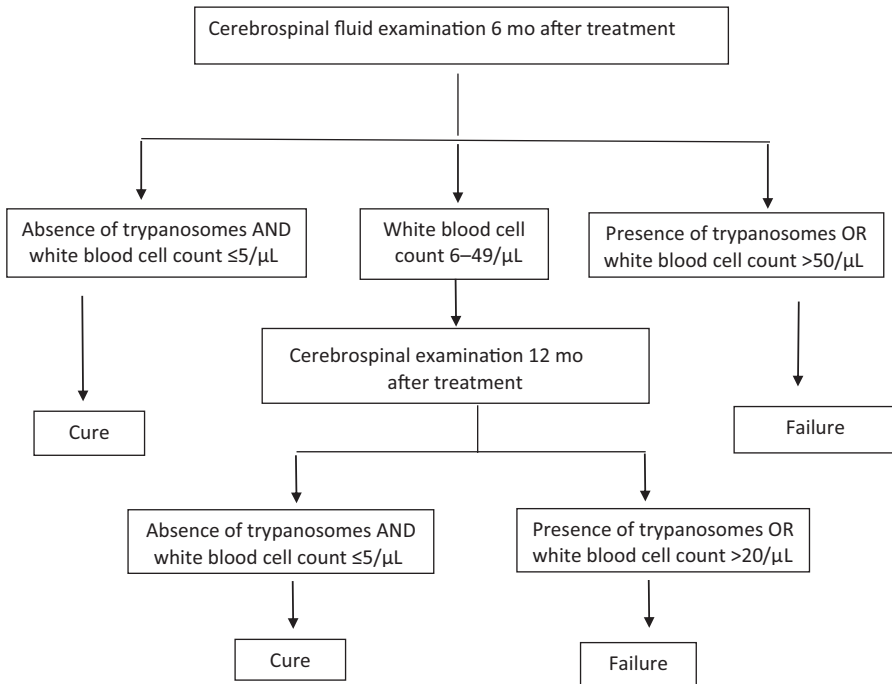


Fig. 1. Proposed algorithm for the follow-up of second-stage *T. brucei gambiense* HAT. (Adapted from Mumba Ngoyi D, Lejon V, Pyana P, et al. How to shorten patient follow-up after treatment for *Trypanosoma brucei gambiense* sleeping sickness. *J Infect Dis* 2010;201(3):453–63. Available at: <http://www.ncbi.nlm.nih.gov/pubmed/20047500> http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=PubMed&dopt=Citation&list_uids=20047500; with permission.)

Treatment of failure or relapse after NECT requires expert advice, because melarsoprol therapy should be considered in such cases.²⁷

***Trypanosoma brucei rhodesiense* Human African Trypanosomiasis Treatment Did Not Progress at All**

Hemolymphatic stage

Suramin is highly effective for the treatment of first-stage infection due to *T brucei rhodesiense* and has been used in the field since 1922.⁵⁶ Usually, the blood smear becomes negative within 3 days after the first dose given. Its long half-life (44–54 days) allows a once-weekly or twice-weekly administration via the intramuscular or intravenous route. A test dose of suramin needs to be administered, because of the risk of anaphylactic reaction. Nephropathy is a potential adverse event, and needs regular monitoring.

Pentamidine, which is more widely available, also has activity against *T brucei rhodesiense*, and could be considered first if the suramin therapy has to be delayed. Some cases have even been successfully treated with pentamidine only.^{17,18} There is no therapeutic advantage of combining suramin and pentamidine.

In travelers who usually present very early after the onset of first-stage HAT symptoms, the systematic CSF examination is debatable because it takes several weeks before trypanosomes invade the CNS. In case the decision is taken to perform a

lumbar puncture in such patients, most experts recommend to do it 2 to 3 days after the administration of the first pentamidine dose, when the parasitemia has become negative; this should avoid contaminating the CSF with blood trypanosomes and making an erroneous diagnosis of second-stage infection.

Meningoencephalitic stage

Eflornithine is not sufficiently active against *T brucei rhodesiense*,⁷⁷ and therefore melarsoprol has remained the only proven therapeutic option for more than 50 years. Regimens, initially long and complex, have been progressively simplified to a 10-day course with no additional toxicity, according to the Improved Application of Melarsoprol (IMPAMEL) III trial.⁷⁸ The same trial could not demonstrate any added value of pretreatment with suramin in second-stage *T brucei rhodesiense* HAT. It remains unclear whether concomitant administration of prednisolone can prevent, at least partly like for *T brucei gambiense* HAT, the encephalitic syndrome, but a reduction of its frequency has been suggested in IMPAMEL III.⁷⁸ The combination of eflornithine and nifurtimox has not been tried so far. There is experimental evidence that fexinidazole has activity against *T brucei rhodesiense*,⁷⁹ but no clinical trial to evaluate it as an alternative treatment has started yet.

The posttreatment follow-up strategy of second-stage *T brucei rhodesiense* HAT is not clearly established, but has been usually modeled after the classic 2-year approach for *T brucei gambiense*.

None of the HAT drugs are commercially available. For drug supply in Europe, contact should be made with Dr J.R. Franco, Control of Neglected Tropical Diseases, WHO (francoj@who.int). In the United States and Canada, HAT treatment is supplied by the Centers for Disease Control and Prevention Drug Service.

SUMMARY

Despite the rather fast-evolving HAT epidemiology, this condition will continue to be occasionally seen in the nonendemic setting, where clinical expertise for this disease is already scarce. Diagnosis and treatment of HAT in travelers have to rely on the slow but undeniable progress achieved in endemic settings, when considering the recent breakthroughs in rapid diagnostic test screening and in simplified therapeutic options. Confirmatory microscopic diagnosis and species-specific molecular identification, however, still require a level of expertise that is available in only a few reference laboratories. Effective and nontoxic drugs are still in development for use in both African populations and in travelers to treat *T brucei rhodesiense* HAT, that until now has to be managed as in the past century.

REFERENCES

1. Büscher P, Cecchi G, Jamonneau V, et al. Human African trypanosomiasis. *Lancet* 2017;390(10110):2397–409.
2. Lejon V, Boelaert M, Jannin J, et al. The challenge of *Trypanosoma brucei gambiense* sleeping sickness diagnosis outside Africa. *Lancet Infect Dis* 2003;3(12):804–8.
3. Lestrade-Carluer de Kyvon MA, Maakaroun-Vermesse Z, Lanotte P, et al. Congenital trypanosomiasis in child born in France to African mother. *Emerg Infect Dis* 2016;22(5):935–7.
4. Gaillot K, Lauvin M-A, Cottier J-P. Vertical transmission of human African trypanosomiasis: clinical evolution and brain MRI of a mother and her son. *PLoS Negl Trop Dis* 2017;11(7):e0005642.

5. Franco JR, Simarro PP, Diarra A, et al. Epidemiology of human African trypanosomiasis. *Clin Epidemiol* 2014;6(1):257–75.
6. Simarro PP, Cecchi G, Paone M, et al. The atlas of human African trypanosomiasis: a contribution to global mapping of neglected tropical diseases. *Int J Health Geogr* 2010;9(1):57. Available at: <http://www.ij-healthgeographics.com/content/9/1/57>.
7. Franco JR, Cecchi G, Priotto G, et al. Monitoring the elimination of human African trypanosomiasis: update to 2014. *PLoS Negl Trop Dis* 2017;11(5):1–26.
8. Aksoy S, Buscher P, Lehane M, et al. Human African trypanosomiasis control: achievements and challenges. *PLoS Negl Trop Dis* 2017;11(4):1–6.
9. Büscher P, Bart JM, Boelaert M, et al. Do cryptic reservoirs threaten gambiense-sleeping sickness elimination? *Trends Parasitol* 2018;34(3):197–207.
10. Neuberger A, Meltzer E, Leshem E, et al. The changing epidemiology of human African trypanosomiasis among patients from nonendemic countries - 1902–2012. *PLoS One* 2014;9(2):1–7.
11. Migchelsen SJ, Büscher P, Hoepelman AIM, et al. Human African trypanosomiasis: a review of non-endemic cases in the past 20 years. *Int J Infect Dis* 2011;15(8):e517–24.
12. Luintel A, Lowe P, Cooper A, et al. Case of Nigeria-acquired human African trypanosomiasis in United Kingdom, 2016. *Emerg Infect Dis* 2017;23(7):1225–6.
13. Gómez-Junyent J, Pinazo MJ, Castro P, et al. Human African trypanosomiasis in a Spanish traveler returning from Tanzania. *PLoS Negl Trop Dis* 2017;11(3):1–5.
14. Streit JA, Matsumoto E. African trypanosomiasis. *N Engl J Med* 2016;375(24):2380.
15. Wang X, Ruan Q, Xu B, et al. Human African trypanosomiasis in emigrant returning to China from Gabon, 2017. *Emerg Infect Dis* 2018;24(2):400–2.
16. Liu Q, Chen X-L, Chen M-X, et al. *Trypanosoma brucei rhodesiense* infection in a Chinese traveler returning from the Serengeti National Park in Tanzania. *Infect Dis Poverty* 2018;7(1):5–10.
17. Simarro PP, Franco JR, Cecchi G, et al. Human African trypanosomiasis in non-endemic countries (2000–2010). *J Travel Med* 2012;19(1):44–53.
18. Clerinx J, Vlieghe E, Asselman V, et al. Human African trypanosomiasis in a Belgian traveller returning from the Masai Mara area, Kenya, February 2012. *Euro Surveill* 2012;17(10):4–7.
19. Wolf T, Wichelhaus T, Göttig S, et al. *Trypanosoma brucei rhodesiense* infection in a German traveller returning from the Masai Mara area, Kenya, January 2012. *Euro Surveill* 2012;17(10):2–4.
20. Gobbi F, Bisoffi Z. Human African trypanosomiasis in travellers to Kenya. *Euro Surveill* 2012;17(10):1–3.
21. Schwartz E, Michaeli S, Steinlauf S, et al. Human African trypanosomiasis in a traveler: diagnostic pitfalls. *Am J Trop Med Hyg* 2012;87(2):264–6. Available at: <http://www.ajtmh.org/content/journals/10.4269/ajtmh.2012.11-0512>.
22. Cottle LE, Peters JR, Hall A, et al. Multiorgan dysfunction caused by travel-associated African trypanosomiasis. *Emerg Infect Dis* 2012;18(2):287–9.
23. Paul M, Stefaniak J, Smuszkievicz P, et al. Outcome of acute East African trypanosomiasis in a Polish traveller treated with pentamidine. *BMC Infect Dis* 2014;14(1):1–8.
24. Denny MC, Lai LL, Laureno R. Human African trypanosomiasis encephalitis in the United States. *Neurohospitalist* 2016;6(4):170–3.
25. Sudarshi D, Lawrence S, Pickrell WO, et al. Human African trypanosomiasis presenting at least 29 years after infection—what can this teach us about the

- pathogenesis and control of this neglected tropical disease? PLoS Negl Trop Dis 2014;8(12):8–12.
26. Leder K, Torresi J, Libman MD, et al. GeoSentinel surveillance of illness in returned travelers, 2007–2011. *Ann Intern Med* 2013;158(6):456–68.
 27. World Health Organization. Control and surveillance of human African trypanosomiasis. *World Health Organ Tech Rep Ser* 2013;(984):1–237.
 28. Checchi F, Filipe JAN, Haydon DT, et al. Estimates of the duration of the early and late stage of gambiense sleeping sickness. *BMC Infect Dis* 2008;8:1–10.
 29. Boa Y, Traore M, Doua F, et al. Les différents tableaux cliniques actuels de la trypanosomiase humaine africaine à *T. b. gambiense*. Analyse de 300 dossiers du foyer de Daloa, Côte d'Ivoire. *Bull Soc Pathol Exot Filiales* 1988;81:427–44.
 30. Eperon G, Schmid C, Loutan L, et al. Clinical presentation and treatment outcome of sleeping sickness in Sudanese pre-school children. *Acta Trop* 2007;101:31–9.
 31. Blum J, Schmid C, Burri C. Clinical aspects of 2541 patients with second stage human African trypanosomiasis. *Acta Trop* 2006;97(1):55–64.
 32. Kennedy PGE. Clinical features, diagnosis, and treatment of human African trypanosomiasis (sleeping sickness). *Lancet Neurol* 2013;12(2):186–94.
 33. Checchi F, Funk S, Chandramohan D, et al. Updated estimate of the duration of the meningo-encephalitic stage in gambiense human African trypanosomiasis. *BMC Res Notes* 2015;8(1):8–10.
 34. Blum JA, Burri C, Hatz C, et al. Sleeping hearts: the role of the heart in sleeping sickness (human African trypanosomiasis). *Trop Med Int Health* 2007;12(12):1422–32.
 35. Blum JA, Schmid C, Hatz C, et al. Sleeping glands? The role of endocrine disorders in sleeping sickness (*T.b. gambiense* human African trypanosomiasis). *Acta Trop* 2007;104(1):16–24.
 36. Urech K, Neumayr A, Blum J. Sleeping sickness in travelers—do they really sleep? *PLoS Negl Trop Dis* 2011;5(11):e1358.
 37. Kato CD, Nanteza A, Mugasa C, et al. Clinical profiles, disease outcome and comorbidities among *T.b. rhodesiense* sleeping sickness patients in Uganda. *PLoS One* 2015;10(2):1–13.
 38. MacLean LM, Odiit M, Chisi JE, et al. Focus-specific clinical profiles in human African trypanosomiasis caused by *Trypanosoma brucei rhodesiense*. *PLoS Negl Trop Dis* 2010;4(12):e906.
 39. Kuepfer I, Hhary EP, Allan M, et al. Clinical presentation of *T.b. rhodesiense* sleeping sickness in second-stage patients from Tanzania and Uganda. *PLoS Negl Trop Dis* 2011;5(3):e968.
 40. Gautret P, Clerinx J, Caumes E, et al. Imported human African trypanosomiasis in Europe, 2005–2009. *Euro Surveill* 2009;14(36):9–11.
 41. Mumba Ngoyi D, Menten J, Pyana PP, et al. Stage determination in sleeping sickness: comparison of two cell counting and two parasite detection techniques. *Trop Med Int Health* 2013;18(6):778–82.
 42. Lejon V, Reiber H, Legros D, et al. Intrathecal immune response pattern for improved diagnosis of central nervous system involvement in trypanosomiasis. *J Infect Dis* 2003;187(9):1475–83. Available at: <http://www.ncbi.nlm.nih.gov/pubmed/12717630>.
 43. Tiberti N, Hainard A, Lejon V, et al. Cerebrospinal fluid neopterin as marker of the meningo-encephalitic stage of *Trypanosoma brucei gambiense* sleeping sickness. *PLoS One* 2012;7(7):e40909.
 44. Tiberti N, Matovu E, Hainard A, et al. New biomarkers for stage determination in *Trypanosoma brucei rhodesiense* sleeping sickness patients. *Clin Transl Med*

- 2013;2(1):1. Available at: <http://clintransmed.springeropen.com/articles/10.1186/2001-1326-2-1>.
45. Chappuis F, Loutan L, Simarro P, et al. Options for field diagnosis of human African trypanosomiasis. *Clin Microbiol Rev* 2005;18(1):133–46.
 46. Mitashi P, Hasker E, Lejon V, et al. Human African trypanosomiasis diagnosis in first-line health services of endemic countries, a systematic review. *PLoS Negl Trop Dis* 2012;6(11):e1919.
 47. Deborggraeve S, Büscher P. Molecular diagnostics for sleeping sickness: what is the benefit for the patient? *Lancet Infect Dis* 2010;10(6):433–9.
 48. Mumba Ngoyi D, Ali Ekangu R, Mumvumba Kodi MF, et al. Performance of parasitological and molecular techniques for the diagnosis and surveillance of gambiense sleeping sickness. *PLoS Negl Trop Dis* 2014;8(6):e2954.
 49. González-Andrade P, Camara M, Ilboudo H, et al. Diagnosis of trypanosomatid infections: targeting the spliced leader RNA. *J Mol Diagn* 2014;16(4):400–4.
 50. Yansouni CP, Bottieau E, Lutumba P, et al. Rapid diagnostic tests for neurological infections in central Africa. *Lancet Infect Dis* 2013;13(6):546–58.
 51. Buscher P, Gilleman Q, Lejon V. Rapid diagnostic test for sleeping sickness. *N Engl J Med* 2013;368(11):1069–70.
 52. Büscher P, Mertens P, Leclipteux T, et al. Sensitivity and specificity of HAT Sero-K-SeT, a rapid diagnostic test for serodiagnosis of sleeping sickness caused by *Trypanosoma brucei gambiense*: a case-control study. *Lancet Glob Health* 2014;2(6):359–63.
 53. Boelaert M, Mukendi D, Bottieau E, et al. A phase III diagnostic accuracy study of a rapid diagnostic test for diagnosis of second-stage human African trypanosomiasis in the Democratic Republic of the Congo. *EBioMedicine* 2017;27:11–7.
 54. Bisser S, Lumbala C, Nguertoum E, et al. Sensitivity and specificity of a prototype rapid diagnostic test for the detection of *Trypanosoma brucei gambiense* infection: a multi-centric prospective study. *PLoS Negl Trop Dis* 2016;10(4):1–16.
 55. Lumbala C, Biéler S, Kayembe S, et al. Prospective evaluation of a rapid diagnostic test for *Trypanosoma brucei gambiense* infection developed using recombinant antigens. *PLoS Negl Trop Dis* 2018;12(3):1–20.
 56. Babokhov P, Sanyaolu AO, Oyibo WA, et al. A current analysis of chemotherapy strategies for the treatment of human African trypanosomiasis. *Pathog Glob Health* 2013;107(5):242–52.
 57. Burri C, Yeramian PD, Allen JL, et al. Efficacy, safety, and dose of pafuramidine, a new oral drug for treatment of first stage sleeping sickness, in a phase 2a clinical study and phase 2b randomized clinical studies. *PLoS Negl Trop Dis* 2016;10(2):1–18.
 58. Pohlig G, Bernhard SC, Blum J, et al. Efficacy and safety of pafuramidine versus pentamidine maleate for treatment of first stage sleeping sickness in a randomized, comparator-controlled, international phase 3 clinical trial. *PLoS Negl Trop Dis* 2016;10(2):1–17.
 59. Blum J, Nkunku S, Burri C. Clinical description of encephalopathic syndromes and risk factors for their occurrence and outcome during melarsoprol treatment of human African trypanosomiasis. *Trop Med Int Health* 2001;6(5):390–400.
 60. Pepin J, Guerin C, Ethier L, et al. Trial of prednisolone for prevention of melarsoprol-induced encephalopathy in gambiense sleeping sickness. *Lancet* 1989;1(8649):1246–50.
 61. Robays J, Nyamowala G, Sese C, et al. High failure rates of melarsoprol for sleeping sickness, Democratic Republic of Congo. *Emerg Infect Dis* 2008;14(6):966–7.

62. Pyana Pati P, Van Reet N, Mumba Ngoyi D, et al. Melarsoprol sensitivity profile of *Trypanosoma brucei gambiense* isolates from cured and relapsed sleeping sickness patients from the Democratic Republic of the Congo. *PLoS Negl Trop Dis* 2014;8(10):5–10.
63. Milord F, Pepin J, Loko L, et al. Efficacy and toxicity of eflornithine for treatment of *Trypanosoma brucei gambiense* sleeping sickness. *Lancet* 1992;340:652–5.
64. Chappuis F, Udayraj N, Stietenroth K, et al. Eflornithine is safer than melarsoprol for the treatment of second-stage *Trypanosoma brucei gambiense* human African trypanosomiasis. *Clin Infect Dis* 2005;41(5):748–51.
65. Pépin J, Khonde N, Maiso F, et al. Short-course eflornithine in Gambian trypanosomiasis: a multicentre randomized controlled trial. *Bull World Health Organ* 2000;78(11):1284–95.
66. Priotto G, Kasparian S, Mutombo W, et al. Nifurtimox-eflornithine combination therapy for second-stage African *Trypanosoma brucei gambiense* trypanosomiasis: a multicentre, randomised, phase III, non-inferiority trial. *Lancet* 2009; 374(9683):56–64.
67. Schmid C, Kuemmerle A, Blum J, et al. In-hospital safety in field conditions of nifurtimox eflornithine combination therapy (NECT) for *T. b. gambiense* sleeping sickness. *PLoS Negl Trop Dis* 2012;6(11):e1920.
68. Alirol E, Schruppf D, Amici Heradi J, et al. Nifurtimox-eflornithine combination therapy for second-stage gambiense human African trypanosomiasis: Médecins Sans Frontières experience in the Democratic Republic of the Congo. *Clin Infect Dis* 2013;56(2):195–203.
69. Mesu VKBK, Kalonji WM, Bardonneau C, et al. Oral fexinidazole for late-stage African *Trypanosoma brucei gambiense* trypanosomiasis: a pivotal multicentre, randomised, non-inferiority trial. *Lancet* 2017;391:144–54.
70. Jacobs RT, Nare B, Wring SA, et al. Scyx-7158, an orally-active benzoxaborole for the treatment of stage 2 human African trypanosomiasis. *PLoS Negl Trop Dis* 2011;5(6):e1151.
71. Chappuis F. Oral fexinidazole for human African trypanosomiasis. *Lancet* 2018; 391(10116):100–2.
72. Mumba Ngoyi D, Lejon V, Pyana P, et al. How to shorten patient follow-up after treatment for *Trypanosoma brucei gambiense* sleeping sickness. *J Infect Dis* 2010;201(3):453–63. Available at: <http://www.ncbi.nlm.nih.gov/pubmed/20047500> http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=PubMed&dopt=Citation&list_uids=20047500.
73. Lejon V, Roger I, Mumba Ngoyi D, et al. Novel markers for treatment outcome in late-stage *Trypanosoma brucei gambiense* trypanosomiasis. *Clin Infect Dis* 2008; 47(1):15–22.
74. Tiberti N, Lejon V, Hainard A, et al. Neopterin is a cerebrospinal fluid marker for treatment outcome evaluation in patients affected by *Trypanosoma brucei gambiense* sleeping sickness. *PLoS Negl Trop Dis* 2013;7(2):1–9.
75. Ilboudo H, Camara O, Ravel S, et al. *Trypanosoma brucei gambiense* spliced leader RNA is a more specific marker for cure of human African trypanosomiasis than *T. b. gambiense* DNA. *J Infect Dis* 2015;212(12):1996–8.
76. Deborggraeve S, Lejon V, Ekangu RA, et al. Diagnostic accuracy of PCR in gambiense sleeping sickness diagnosis, staging and post-treatment follow-up: a 2-year longitudinal study. *PLoS Negl Trop Dis* 2011;5(2):1–7.
77. Clerinx J, Taelman H, Bogaerts J, et al. Treatment of late stage rhodesiense trypanosomiasis using suramin and eflornithine: report of six cases. *Trans R Soc Trop Med Hyg* 1998;92(4):449–50.

78. Kuepfer I, Schmid C, Allan M, et al. Safety and efficacy of the 10-day melarsoprol schedule for the treatment of second stage rhodesiense sleeping sickness. *PLoS Negl Trop Dis* 2012;6(8):e1695.
79. Kaiser M, Bray MA, Cal M, et al. Antitrypanosomal activity of fexinidazole, a new oral nitroimidazole drug candidate for treatment of sleeping sickness. *Antimicrob Agents Chemother* 2011;55(12):5602–8.